

Six-row trace excess, direct source action, and the correlation boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-30 · this version issued 2026-07-30 · v1.0

1. Purpose and scope

R-122 left the A13 proof at a precise fork. One route tried to cancel both low chaoses of every adapted root and then exponentiate a double divergence. The other route retained the complete signed endpoint and asked only for the expected R-093 source action. Ten proof rounds were used to compare these routes after reconstructing the fixed six-row coefficient law and retaining all owner multiplicities.

R-123 conclusion. Conditionally on the strict past, let Φ be a fully assembled contraction-closed six-row endpoint output, let Θ be its once-owned complete covariance trace, let $b = \mathbb{E}_0\Phi$, and put $Y = \Phi - b$. The one scalar mismatch

$$\Lambda = \Theta - \|Y\|^2$$

contains both R-122 low-chaos defects:

$$D_0 = \mathbb{E}_0\Lambda, \quad D_1 = \mathbb{E}_0[\xi\Lambda].$$

More importantly, the direct complete packet obeys

$$\mathbb{E}_0\mathcal{P}_{\text{comp}} = \frac{1}{2}\|b\|^2 - \frac{1}{2}D_0.$$

Thus the direct R-093 expected source-action route does not require $D_1 = 0$. Its exact irreducible target is one aggregate upper bound for $D_0 - \|b\|^2$ after all six rows, roots, visits, traces, low terms, score terms, rational terms, and R-063 forest owners have been recombined.

A bounded full-six-row finite-root fixture has nonzero D_0 and D_1 but a strictly positive direct packet. A genuine legal first-linear-row fixture admits an expectation bound from source L^2 and one sixth moment with no feedback derivatives and no fifth coefficient moment. These are real advances, but they are not the full production estimate.

Three shortcuts are retired. Stationary six-row parity does not transfer to arbitrary adapted controls. The isolated raw six-current Hessian is indefinite. Correlation-before-Holder removes an exponential amplitude loss, but without spatial gain it cannot provide arbitrary cutoff-uniform Young allocations. The complete cutoff/chart/control-uniform trace-excess estimate, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector-A closure remain open. Tier stays T4.

The result is expectation-level and finite-cutoff. A conditional log-normalizer, determinant, or pure double-divergence route still needs the first-chaos information and the relevant domain hypotheses. No formula below permits rowwise positive parts, duplicated forests, or per-chart constants.

2. Round one: fixed six-row endpoint law and owner incidence

Let $S_A = S_A^*$, $A = 1, 2, 3$, be the fixed six-real Pauli generators. For $w \in \mathbb{R}^6$, a positive density floor e , and a tangent v , write

$$\rho(w) = |w|^2, \quad m_A(w) = w^T S_A w, \quad q_A(w) = \frac{m_A(w)}{\rho(w) + e}, \quad (2.1)$$

$$P_A(w; v) = 2(S_A w)^T v, \quad L_A(w; v) = 2\{(S_A - \alpha q_A(w)I)w\}^T v, \quad (2.2)$$

where the A1 manifest gives

$$P = M_X^2 + \varepsilon_m = 4 + 10^{-12}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9}. \quad (2.3)$$

Define the fixed orthogonal six-row map

$$C_6(w)v = (\sqrt{c_0}P_A(w; v), \sqrt{c_1}L_A(w; v))_{A=1}^3. \quad (2.4)$$

Then the complete raw covariance coefficient is exactly

$$B(w) = C_6(w)^T C_6(w) = \sum_{A=1}^3 \{c_0 p_A p_A^T + c_1 \ell_A \ell_A^T\}. \quad (2.5)$$

This is the R-120 fixed coordinate. A state-dependent rank compression may be used only after the conditional covariance has been formed; it is not a replacement for (2.4).

On the legal R-119 two-block chart set

$$Z_* = S_1(\xi + h_1) + S_2(\eta + h_2(\xi)), \quad W_r = Z_* + r, \quad V_i = \partial_i Z_*. \quad (2.6)$$

For a contraction-closed output cluster $\mathcal{C}$, the heat coordinate r stays inside the output Hilbert space and the six-row endpoint is

$$\Phi_{h,\mathcal{C}}(\xi) = \left(\Pi_{\mathcal{C}} \mathbb{E}_{\eta} [C_6(\widehat{W_r}) V_i(n) \mid \xi] \right)_{i,n \in \mathcal{C}}. \quad (2.7)$$

The covariance-normal trace is

$$\Theta_{h,\mathcal{C}}(\xi) = \mathbb{E}_{\eta, G_1} \sum_{i,n \in \mathcal{C}} \int \left| \Pi_{\mathcal{C}} C_6(\widehat{W_r}) \partial_i G_1(n) \right|^2 \mu_{\Sigma}(dr) \mid \xi. \quad (2.8)$$

Here G_1 is the fresh covariance tangent. Formula (2.8) is the primitive trace owner. The R-063 Wick forest is an alternative expansion of the same endpoint reconstruction and must not be added a second time.

The owner-incidence audit used in all later rounds is:

owner family	net incidence	
current/nonlinear cross	+1	
nonlinear square	+1/2	
safe covariance trace	-1/2	
affine Hilbert-Schmidt rebate	+1/2	
heat internal expansion or R-063 alternative forest	0 extra	
intermediate visit endpoint	0	(2.9)
$R_Q + M_U + K_R$ rational aggregate	1 total	
conditional low and complete low	1 each	
paid difference	1	
Cartan FAR	1 REG copy	
duplicate q/r , Schur, R-076, or R-086 payment	0.	

The table is an ownership firewall, not a positivity statement.

3. Round two: scalar trace-excess collapse

Condition on the strict past $\mathcal{G}$ and write $\mathbb{E}_0 = \mathbb{E}[\cdot \mid \mathcal{G}]$. For one fully owned cluster define

$$b = \mathbb{E}_0 \Phi, \quad Y = \Phi - b, \quad A = \mathbb{E}_0[Y \otimes \xi], \quad R = Y - A\xi. \quad (3.1)$$

Thus R is orthogonal to zero and first Gaussian chaos. R-122 wrote

$$D_0 = \mathbb{E}_0 \Theta - \|A\|_{\text{HS}}^2 - \mathbb{E}_0 \|R\|^2, \quad (3.2)$$

$$D_1 = \mathbb{E}_0[\xi \Theta] - 2 \text{Contr}(A, \mathbb{E}_0[R \otimes H_2(\xi)]) - \mathbb{E}_0[\xi \|R\|^2]. \quad (3.3)$$

Since

$$\mathbb{E}_0 \|Y\|^2 = \|A\|_{\text{HS}}^2 + \mathbb{E}_0 \|R\|^2, \quad (3.4)$$

and

$$\mathbb{E}_0[\xi \|Y\|^2] = 2 \text{Contr}(A, \mathbb{E}_0[R \otimes H_2(\xi)]) + \mathbb{E}_0[\xi \|R\|^2], \quad (3.5)$$

the derivative-free scalar

$$\boxed{\Lambda = \Theta - \|Y\|^2} \quad (3.6)$$

gives the exact collapse

$$\boxed{D_0 = \mathbb{E}_0 \Lambda, \quad D_1 = \mathbb{E}_0[\xi \Lambda]}. \quad (3.7)$$

No Dh , D^2h , or termwise Hermite expansion appears in (3.7).

Let (ξ', Φ', Θ') be a conditionally independent copy and let $\Lambda' = \Theta' - \|\Phi' - b\|^2$. Then

$$\boxed{D_0 = \mathbb{E}_0 \Theta - \frac{1}{2} \mathbb{E}_0 \|\Phi - \Phi'\|^2}, \quad (3.8)$$

$$\boxed{D_1 = \frac{1}{2} \mathbb{E}_0[(\xi - \xi')(\Lambda - \Lambda')]}. \quad (3.9)$$

Thus D_0 is a conditional trace excess over replica variance, while D_1 is the root covariance of that mismatch.

4. Round three: direct action removes the first-chaos burden

The complete square-minus-trace packet is

$$\mathcal{P}_{\text{comp}} = \frac{1}{2} \|\Phi\|^2 - \frac{1}{2} \Theta. \quad (4.1)$$

Using $\Phi = b + Y$, $\mathbb{E}_0 Y = 0$, and (3.7),

$$\boxed{\mathbb{E}_0 \mathcal{P}_{\text{comp}} = \frac{1}{2} \|b\|^2 - \frac{1}{2} D_0 = -\frac{1}{2} \mathbb{E}_0[\Theta - \|\Phi\|^2]}. \quad (4.2)$$

The load-bearing scalar is therefore

$$\boxed{D_0 - \|b\|^2 = \mathbb{E}_0 \Theta - \mathbb{E}_0 \|\Phi\|^2}. \quad (4.3)$$

The positive $\|b\|^2$ reserve cannot be discarded. Later feedback is not being ignored: it must already be inside the contraction-closed Φ and once-owned Θ before (4.2) is used.

For comparison, the R-119 coherent residual has

$$\Pi_0 \mathcal{R}_{\text{coh}} = -\frac{1}{2} D_0, \quad \Pi_1 \mathcal{R}_{\text{coh}} = -\frac{1}{2} \langle D_1, \xi \rangle. \quad (4.4)$$

For finite Gaussian polynomials, or under a separately proved closed double-divergence domain,

$$\mathcal{R}_{\text{coh}} = -\frac{1}{2} D_0 - \frac{1}{2} \langle D_1, \xi \rangle + \frac{1}{2} \delta^2 W_{\text{can}}. \quad (4.5)$$

The coefficient W_{can} may change sign. Hence (4.5) is useful for bookkeeping but supplies neither a positive coefficient nor an exponential estimate. The direct expectation (4.2) is strictly weaker and needs no $D_1 = 0$.

5. Round four: symmetry, exchange, and trace substitutes

The weakest root reflection assumption that kills the first defect is

$$\Lambda(-\xi) = \Lambda(\xi). \quad (5.1)$$

It implies $D_1 = 0$ but says nothing about D_0 . The R-120 stationary theorem has the stronger mechanism: conditionally independent centered Gaussian value and derivative, plus fixed-row parity, give both the variance identity and reflection. Same-root adapted feedback destroys those hypotheses, so the stationary argument cannot be transferred to the R-093 source union.

Canonical variance representatives also fail to reproduce both low chaoses in general. For a real scalar finite polynomial

$$Y = \sum_{n \geq 1} c_n H_n(G), \quad \mathbb{E} H_n(G)^2 = n!, \quad (5.2)$$

the following table gives the mean weight on $n!c_n^2$ and the first-chaos weight on $n!c_n c_{n+1}$:

representative	mean weight	first weight
$\frac{1}{2} \mathbb{E}'[(Y - Y')^2 \mid G]$	1	$n + 1$
Clark–Ocone bracket, $G = B_1$	1	$2n$
$\langle DY, -DL^{-1}Y \rangle$	1	$2n + 1$
Θ_{sc}	1	$2(n + 1)$
$\ DY\ ^2$	n	$2n(n + 1)$.

(5.3)

The table is one-dimensional and uses probabilists' unnormalised Hermites. The Clark row depends on the displayed Brownian realisation. It is a method audit, not a production trace identity.

For $Y = \varepsilon(aH_1 + H_2)$, the required pair is

$$(\text{mean}, \text{first}) = \varepsilon^2(a^2 + 2, 4a). \quad (5.4)$$

Exchange and Clark give first coefficient $2a$, the OU trace gives $3a$, the score-compatible trace gives $4a$, and the raw derivative gives mean $a^2 + 4$. No canonical substitution matches both entries without an additional production identity.

The raw derivative mean debt has the exact OU-replica formula

$$\mathbb{E} \|DY\|^2 - \text{Var}(Y) = \frac{1}{2} \int_0^1 \mathbb{E} \|DY(G) - DY(G_\rho)\|^2 d\rho = \sum_{n \geq 2} (n-1) n! c_n^2, \quad (5.5)$$

where $G_\rho = \rho G + \sqrt{1-\rho^2} G'$. R-122's $h_N(G) = \sin(NG)/N$ makes its value and sixth-moment budgets vanish while the derivative defect stays of order one. A derivative-secant replacement therefore cannot close A13.

6. Round five: a bounded full-six-row noncancellation fixture

Let ξ, ζ be independent standard Gaussians and take the active real doublet fixture

$$A(\xi) = a + \beta \sin(t\xi), \quad w = (A(\xi) + \tau\zeta)e_1, \quad y = \xi e_2. \quad (6.1)$$

Only the S_1 pair in (2.4) survives, with $q_1 = 0$ and $L_1 = P_1$. Put

$$s = c_0 + c_1 = \frac{339}{8000P}. \quad (6.2)$$

Then

$$\Phi = 2(A + \tau\zeta)\xi(\sqrt{c_0}, \sqrt{c_1}), \quad \Theta = 4s(A^2 + \tau^2). \quad (6.3)$$

Gaussian characteristic differentiation gives

$$\|b\|^2 = 4s\beta^2 t^2 e^{-t^2}, \quad (6.4)$$

$$D_0 = 4s\beta^2 t^2 (e^{-t^2} - 2e^{-2t^2}), \quad (6.5)$$

$$D_1 = 8sa\beta t(t^2 - 1)e^{-t^2/2}, \quad (6.6)$$

and nevertheless

$$\boxed{\mathbb{E}\mathcal{P}_{\text{comp}} = 4s\beta^2 t^2 e^{-2t^2} > 0} \quad (6.7)$$

when $\beta t \neq 0$. At $t = 2$,

$$D_0 = 16s\beta^2(e^{-4} - 2e^{-8}), \quad D_1 = 48sa\beta e^{-2}, \quad \mathbb{E}\mathcal{P}_{\text{comp}} = 16s\beta^2 e^{-8}. \quad (6.8)$$

The independent heat cancels from (6.4)–(6.8). Thus automatic adapted D_0, D_1 cancellation is false even in the fixed six-row coordinate, but nonzero defects do not imply a bad direct packet. This is a bounded finite-root coefficient/derivative fixture, not a spatial Cameron–Martin production theorem.

7. Round six: exact legal first-linear-row expectation theorem

For the R-122 normalized circle fixture and any strict-past scalar control $h(\xi)$, the complete first-linear-row output is

$$\Phi_h = \kappa \left[\frac{\xi^2}{\sqrt{2}} e_{s2} + \xi h(\xi) e_c \right], \quad \Theta_h = \kappa^2 \left(\frac{\xi^2}{2} + h^2 + 1 \right), \quad (7.1)$$

with

$$\kappa^2 = 4c_0 = \frac{6}{125P}. \quad (7.2)$$

The exact pointwise packet is

$$\mathcal{P}_{\text{row}} = \frac{\kappa^2}{4}(\xi^4 - \xi^2 - 2) + \frac{\kappa^2}{2}(\xi^2 - 1)h^2. \quad (7.3)$$

The first polynomial has zero expectation but may not be deleted inside an exponential. At expectation level,

$$\boxed{\mathbb{E}\mathcal{P}_{\text{row}} = \frac{\kappa^2}{2}\mathbb{E}[(\xi^2 - 1)h^2]}. \quad (7.4)$$

For smooth C^2 data with the displayed integrability, Gaussian integration by parts gives the equivalent score form

$$\boxed{\mathbb{E}\mathcal{P}_{\text{row}} = \kappa^2\mathbb{E}\{(h')^2 + hh''\}}. \quad (7.5)$$

The derivative-free form (7.4) needs only $h \in L^6(\gamma)$; this also gives $\xi h \in L^2$.

The exact mean debt is

$$D_0(h) = \kappa^2 \left\{ \frac{1}{2} + \mathbb{E}h^2 - \text{Var}(\xi h) \right\}, \quad (7.6)$$

so

$$D_0(h) - D_0(0) \leq \kappa^2 \mathbb{E}h^2, \quad \kappa^2 \simeq 0.012 < \frac{9}{20}. \quad (7.7)$$

This closes the mean debt of this one genuine adapted row from source cost, without Dh , D^2h , D_1 , or a fifth coefficient moment.

R-109 with fixed scalar $W = 1$ gives, for $X = \mathbb{E}h^2$ and $Y = \mathbb{E}|h|^6$,

$$|\mathbb{E}\mathcal{P}_{\text{row}}| \leq \frac{\kappa^2}{\sqrt{2}} X^{1/4} Y^{1/4}. \quad (7.8)$$

Therefore for every $\eta, \zeta > 0$,

$$\boxed{\mathbb{E}\mathcal{P}_{\text{row}} \geq -\eta X - \zeta Y - \frac{\kappa^4}{16\sqrt{\eta\zeta}}.} \quad (7.9)$$

The denominator is 16. R-109's denominator 64 applies to $-q\mathfrak{F}_W/2$, not to $\kappa^2\mathfrak{F}_1$ in (7.5).

For $h_{\pm} = a \sin \xi \pm d \cos \xi$,

$$\mathbb{E}\mathcal{P}_{\text{row},\pm} = \kappa^2 e^{-2}(a^2 - d^2), \quad D_{1,\pm} = \pm 4\kappa^2 a d e^{-2}. \quad (7.10)$$

The direct expectation ignores the sign which flips D_1 . Identifying the single Y in (7.9) with the canonical spatial terminal sextic, and summing the constant without shell or visit multiplicity, remain production duties.

8. Round seven: multiplicity-free directed-union criterion

R-107 gives the exact expected source action on each temporally faithful chart:

$$\mathcal{A}_J(h) = \sum_{k,\mathcal{C}} \mathbb{E}\mathcal{P}_{k,\mathcal{C}}^{\text{comp}} + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 + \frac{3}{20} \mathbb{E} \|Z_K\|_6^6. \quad (8.1)$$

Consequently the aggregate hypothesis

$$\boxed{\sum_{k,\mathcal{C}} \mathbb{E}[D_{0,k,\mathcal{C}} - \|b_{k,\mathcal{C}}\|^2] \leq 2\eta \mathbb{E} \sum_k \|h_k\|^2 + 2\zeta \mathbb{E} \|Z_K\|_6^6 + 2C} \quad (8.2)$$

implies

$$\boxed{\mathcal{A}_J(h) \geq \left(\frac{9}{20} - \eta \right) \mathbb{E} \sum_k \|h_k\|^2 + \left(\frac{3}{20} - \zeta \right) \mathbb{E} \|Z_K\|_6^6 - C.} \quad (8.3)$$

Here b may be strict-past random, so the outer expectation in (8.2) is essential. The useful range is $\eta < 9/20$ and $\zeta < 3/20$.

Let $\mathfrak{T}_J$ be R-093's directed union of canonical temporally faithful source charts. If (8.3) holds with the same C, η, ζ for every chart and control, taking the infimum over $\mathfrak{T}_J$ creates no chart-count factor. This does not remove multiplicity already created inside a chart. The safe uniformity checklist is:

1. recombine every row, root, visit, and owner before absolute values;
2. require invariance under representation-preserving subdivision;

3. require the endpoint form to descend through the synthesis kernel, for example $K_\pi = L_\pi^* B_\pi L_\pi$;
4. use R-118 absolute spectral variation, or R-120 multiplier derivatives through spatial order two, not operator norm alone;
5. keep the H^2 synthesis, endpoint L^6 bridge, rough $10/3$ moment, and Gaussian baseline uniform in cutoff and chart;
6. forbid constants per row, root, visit, block, or refinement depth;
7. retain strict-past auxiliary information and legal heat disintegration;
8. distinguish the fixed- J R-093 equality from the separate cutoff-uniform Nelson bound.

No chart need attain the infimum. A uniform lower bound for every chart passes directly to it.

9. Round eight: exact counterexamples and correlation boundary

9.1 Raw six-current Hessian positivity is false

On the active real doublet use normalized Haar measure and unit phase frequency, and set

$$u_H(x) = H(2 + \cos x)e_1, \quad z(x) = (2 - \cos x)e_1. \quad (9.1)$$

All off-diagonal Pauli rows vanish. The surviving diagonal linear/rational pair has energy

$$E_{\text{cur}}(U) = 2 \int U^2 \left[c_0 + c_1 \left(1 - \frac{5}{9} \frac{U^2}{U^2 + e} \right)^2 \right] (U')^2 dx. \quad (9.2)$$

Since

$$c_0 + c_1(4/9)^2 = \frac{9}{500P}, \quad (9.3)$$

and

$$\left\langle (6 \cos^2 x - 8) \sin^2 x \right\rangle = -\frac{13}{4}, \quad (9.4)$$

one obtains

$$\boxed{D^2 E_{\text{cur}}(u_H)[z, z] = -\frac{117}{500P} H^2 + \frac{3e}{100P} + O_e(H^{-2}) < 0} \quad (9.5)$$

for large finite H . With the repository's unnormalised $\mathbb{T}_L^3$ integral and a legal mode $k = 2\pi n/L$, multiply (9.5) by $L^3|k|^2$. This disproves raw-current/phase-pair Hessian positivity. It is not a counterexample to the complete trace/heat/low/forest/sextic action.

9.2 Correlation eliminates an exponential Holder loss

For the R-122 coherent amplitude

$$A_t = e^{t\xi - 3t^2}, \quad (9.6)$$

one has

$$\mathbb{E} A_t^p = e^{p(p-6)t^2/2}. \quad (9.7)$$

Coefficient-first extraction of a quadratic current in L^5 requests

$$\mathbb{E} A_t^{10} = e^{20t^2}, \quad (9.8)$$

whereas the actual correlated quadratic-coefficient/quadratic-payload product uses

$$\mathbb{E} A_t^4 = e^{-4t^2}. \quad (9.9)$$

Thus separated Holder loses the exact factor e^{24t^2} . Smooth bounded caps preserve the separation. This proves that correlation-first assembly is materially sharper, not that it closes the spatial theorem.

9.3 Fixed-profile Young is not cutoff-uniform

For a fixed skew matrix J and

$$z_{A,N} = A(\cos Nx, \sin Nx), \quad z_{A,N}^T J \partial_x z_{A,N} = -A^2 N, \quad (9.10)$$

the correlated quartic scale is cA^4N^2 , while the available model scales are

$$X \asymp A^2N^4, \quad Y = A^6. \quad (9.11)$$

A uniform estimate

$$cA^4N^2 \leq \eta A^2N^4 + \zeta A^6 + C \quad (9.12)$$

forces

$$\boxed{\eta\zeta \geq c^2/4.} \quad (9.13)$$

Indeed set $A^2 = kN^2$, divide by kN^6 , minimise $\eta/k + \zeta k = 2\sqrt{\eta\zeta}$, and let $N \rightarrow \infty$. Therefore arbitrary small allocations still require a production spatial/root gain, a sufficiently large fixed budget product, or a signed cancellation. The circle is a fixed-skew diagnostic, not an A1 or Nelson counterexample.

10. Round nine: strongest valid theorem after ten-route synthesis

The preceding rounds yield the following exact conditional theorem.

Theorem 10.1 (complete trace-excess reduction)

At fixed cutoff and positive floor, assume that one legal strict-past root cluster has been assembled into a contraction-closed Φ and a once-owned Θ , with all later-feedback and R-063 owners included exactly once. Then (3.7)–(4.3) hold. In particular, its direct expected contribution is bounded below if and only if its signed trace excess $\mathbb{E}_0\Theta - \mathbb{E}_0\|\Phi\|^2$ is bounded above. No first-chaos cancellation is needed for this expectation statement.

Corollary 10.2 (uniform directed-union sufficient condition)

If the fully aggregated production family satisfies (8.2) with constants uniform in cutoff, chart, control, dimension, visit count, block count, and representation-preserving refinement, with $\eta < 9/20$ and $\zeta < 3/20$, then (8.3) holds uniformly and the R-093 infimum has a finite lower bound. This is exactly the remaining Nelson-level burden, not a proof that its hypothesis holds.

Corollary 10.3 (one legal adapted row)

For the first-linear-row fixture (7.1), (7.4), (7.6), and (7.9) hold. This one row is paid at expectation level from source L^2 and one sixth moment. It supplies a successful prototype for the direct route but does not identify the complete rational/Cartan/forest aggregate.

The irreducible successor is now smaller than the R-122 target:

$$\boxed{\text{prove (8.2) for the stationary-subtracted, owner-complete six-row rational/Cartan/forest aggregate.}} \quad (10.1)$$

Computing or cancelling D_1 is deferred unless the direct expected-action route fails and a conditional exponential method becomes necessary.

11. Round ten: Devil’s-advocate review and proof-roadmap map

1. **Objection:** $D_1 = 0$ is always necessary. **DISMISSED** for direct expectation; **VALID** for stronger routes. Equation (4.2) removes it after complete ownership. Determinant and pure double-divergence routes still see (4.4).
2. **Objection:** bounding D_0 alone is enough. **UPHELD AGAINST THE CLAIM.** The exact target is $D_0 - \|b\|^2$. The bounded six-row fixture (6.8) shows why the mean reserve is load-bearing.
3. **Objection:** the six rows may be bounded separately and summed. **UPHELD AGAINST THE CLAIM.** Rowwise positive parts and Young inequalities discard signed cross-owner reserves and can introduce row, visit, or shell multiplicity. Equations (2.9) and (8.2) apply only after complete aggregation.
4. **Objection:** R-120 stationary parity proves the adapted result. **UPHELD AGAINST THE CLAIM.** Same-root feedback destroys value–derivative independence and can mix parity. Equations (6.5)–(6.6) give an exact bounded noncancellation fixture.
5. **Objection:** correlation-first Holder closes the spatial bound. **UPHELD AGAINST THE CLAIM.** It removes the exact e^{24t^2} loss, but (9.13) proves that arbitrary cutoff-uniform allocations still need spatial gain or cancellation.
6. **Objection:** (9.5) refutes A13. **DISMISSED.** It refutes only positivity of the isolated raw-current Hessian. The complete trace, heat, low, forest, and sextic action was not inserted.
7. **Objection:** the R-093 directed union removes every multiplicity. **UPHELD AGAINST THE CLAIM.** It removes chart-count multiplicity only after one chart-independent estimate. It cannot repair losses already created inside a chart or a constant that grows with refinement.
8. **Objection:** a fixed-cutoff bound is Nelson. **UPHELD AGAINST THE CLAIM.** R-093 identifies the fixed- J infimum. Uniformity in J is a separate required theorem.

The ten-round evidence map is:

round	route verdict	reusable endpoint	
1	advanced	fixed six-row law and owner firewall	
2	advanced	Λ collapses D_0, D_1	
3	advanced	direct action uses $D_0 - \ b\ ^2$	
4	mixed	symmetry and canonical traces are scoped only	
5	advanced	bounded six-row noncancellation, positive packet	(11.1)
6	advanced	one legal row paid from L^2/L^6	
7	advanced	uniform chart infimum has no chart-count factor	
8	failed shortcuts	raw Hessian and fixed-profile Young no-gos	
9	advanced	trace-excess theorem and exact sufficient criterion	
10	scoped	stationary-subtracted complete aggregate remains open.	

The proof route should not return to coefficient-first fifth moments, automatic stationary parity, raw Hessian positivity, derivative-secant closure, or per-chart budget summation unless a stated hypothesis has changed.

12. Result footer

The primary executable derives the A1 coefficients from the production manifest and checks the scalar collapse, action constants, full-six-row fixture, arbitrary-control row, five trace representatives, OU replica

debt, raw-Hessian coefficient, lognormal exponents, frequency boundary, and scope flags. The non-importing standard-library executable independently uses Gaussian Simpson quadrature, torus quadrature, finite differences, and exact rational coefficient arithmetic. The integrated verifier reruns both, checks their stored JSON artifacts, pins all source/note/PDF hashes, and checks the claim, result, failure, exploration, roadmap, task, and changelog surfaces.

R-123 does not prove (8.2) for the full production aggregate. It proves no $\text{OVERLAP}_{\text{src}}$, Nelson estimate, cutoff or floor removal, interacting measure, Sector-A closure, or T5–T7 promotion.

Result ID: A13-CLASSII-SIX-ROW-TRACE-EXCESS-DIRECT-ACTION-CORRELATION-BOUNDARY (R-123).
 Precise statement: Eqs. (2.1)–(2.9), Thm. 10.1, Cor. 10.2, Cor. 10.3, and the exact boundaries in Secs. 5, 6, and 9.
 Scope: fixed finite cutoff and positive rational floor; exact conditional endpoint and bounded finite-root statements; complete uniform production trace-excess estimate open.
 Dependencies: A1; R-063, R-083, R-093, R-103, R-104, R-107, R-109, R-118, R-119, R-120, R-121, R-122.
 Evidence grade: ANALYTIC, EXACT, EXECUTED.
 Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_six_row_trace_excess_direct_action_boundary_verify.py
 Expected output: R-123 integrated PASS; primary and independent child counts, all hashes, PDF, exploration, registry, and surface checks pass.
 Falsification gate: any failed coefficient, trace-excess, direct-action, six-row fixture, row, Hermite, Hessian, correlation, directed-union, authority, PDF, or public-surface check.
 Tier before / after: T4 / T4.
 No-overclaim statement: the complete cutoff-uniform production estimate, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, measure, and Sector A remain open.
 Next required action: stationary-subtract the owner-complete six-row rational/Cartan/forest aggregate and prove Eq. (8.2) once.